

A route to the thermodynamics of colloid-polymer mixtures from structural information

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Liquid-state theory provides a fundamental link between microscopic interactions and macroscopic thermodynamic properties of fluids. Here, we develop a systematic theoretical framework in which thermodynamic properties are obtained directly from the fluid correlation functions, i.e. from either the radial distribution functions or the fluid state static structure factors. Applying our approach to colloid–polymer mixtures, we derive a consistent description that naturally incorporates colloidal excluded-volume effects. Many-body effects arising from polymer-mediated depletion and confinement are accounted for through an effective excluded-volume contribution combined with a direct-correlation-function-based treatment of the attractive interactions. The resulting free-energy yields thermodynamic predictions in quantitative agreement with molecular simulation data.

I. INTRODUCTION

Colloid–polymer mixtures are prototypical soft-matter systems and are ubiquitous in both industrial formulations and biological materials, where they contribute to the organization of complex living structures [1, 2]. Owing to their tunable effective interactions and rich phase behaviour [3–5], they have enabled a wide range of applications in biotechnology and environmental engineering, including drug delivery, biomolecule purification, and water treatment via aqueous two-phase extraction [6, 7]. Among the diverse phenomena exhibited by these systems, liquid–liquid phase separation of polymers in the presence of colloids is of particular interest [8]. The resulting partitioning of colloids between coexisting polymer-rich phases underlies separation processes and has also been implicated in intracellular compartmentalization and other biological phase-separated environments [9]. Understanding these systems therefore requires an accurate description of their thermodynamics, including phase stability, colloidal partitioning, osmotic pressure, and interfacial properties such as wetting and surface tension [10–12].

Over the past several decades, liquid-state theory has established a rigorous framework for relating the microscopic structure of fluids to their macroscopic thermodynamic properties. Central to this description are the radial distribution function (RDF), $g(r)$, the static structure factor, $S(k)$, and the pair direct correlation function (DCF), $c^{(2)}(r)$, which characterize equilibrium fluid structure and correlations [13]. Thermodynamic properties can be obtained from these quantities through well-established routes such as the virial and compressibility equations, while the DCF plays a central role in classical density functional theory (cDFT) [14, 15], providing the link between microscopic correlations and free-energy

functionals [16–18]. Consequently, cDFT has become one of the principal theoretical frameworks for predicting phase equilibria and interfacial properties of inhomogeneous fluids. In practice, however, obtaining accurate thermodynamic properties generally requires evaluating structural correlations over a range of thermodynamic state points together with thermodynamic integration, making such calculations computationally demanding [13]. An alternative strategy is to reconstruct effective pair interaction potentials from structural information and subsequently determine the thermodynamics from the resulting interactions. At low densities, the potential of mean force, $\phi(r) = -k_B T \ln g(r)$, provides an estimate of the pair interaction, but this approximation neglects many-body correlations and therefore breaks down at finite densities [13]. More sophisticated inverse methods, such as inverse Boltzmann iteration (IBI), recover effective interaction potentials from RDFs at finite density [19]; however, the resulting potentials generally retain a dependence on the thermodynamic state point.

These considerations motivate the inverse framework developed in the present work, in which thermodynamic properties are obtained directly from structural correlations rather than reconstructed interaction potentials. The approach requires only either the RDF, or equivalently the static structure factor, at a limited number of thermodynamic state points. We present two complementary routes. The first determines the thermodynamics directly from the RDF, while the second is based on the DCF, which is obtained from the structure factor through the Ornstein–Zernike (OZ) relation [13], $S(k) = [1 - \rho \hat{c}(k)]^{-1}$, where $\hat{c}(k)$ denotes the Fourier transform of the DCF. The resulting free-energy functional can then be employed within cDFT to predict phase behaviour and interfacial properties without the extensive thermodynamic integrations required by conventional approaches or explicit knowledge of the underlying interaction potentials.

To demonstrate the generality of the framework, we

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first apply it to several one-component model colloidal fluids before extending it to colloid-polymer mixtures. In the latter case, we combine a DCF-based treatment of attractive interactions with an effective description of excluded-volume effects arising from the colloids, yielding a consistent free-energy functional for mixtures with interacting polymers. We validate the approach by comparison with available theoretical and simulation data and finally apply it to determine the phase behaviour of colloid-polymer mixtures exhibiting both polymer excluded-volume interactions and attractive polymer-colloid interactions.

II. THEORY

The present formalism is developed for mixtures; however, for clarity we first consider a one-component system. The goal is to develop a framework for determining the thermodynamic properties of a system, including its bulk phase behaviour, from $g(r)$ (or, equivalently, $S(k)$) evaluated at only a limited number of thermodynamic state points.

Given $g(r)$, we distinguish two broad classes of interactions. In systems with purely soft interactions, such as polymeric systems where the soft ‘particles’ can interpenetrate, $g(r)$ remains finite over all relevant length scales [5]. In contrast, systems with excluded-volume interactions exhibit a finite distance σ_u such that $g(r) = 0$ for $r < \sigma_u$, where σ_u defines an effective core diameter. In these cases, the interaction potential is assumed to have a decomposable form given as,

$$\phi(r) = \phi^{\text{ref}}(r) + \phi^{\text{tail}}(r), \quad (1)$$

where $\phi^{\text{ref}}(r)$ describes the short-range repulsive core and $\phi^{\text{tail}}(r)$ represents the long range attractive (or soft repulsive) contribution and remains finite at all separations. We assume that the contribution to the thermodynamics from the core can be treated as hard-spheres, with a suitable choice of effective diameter $\sigma_{\text{opt}} \neq \sigma_u$ [13, 20–22]. A novelty of our approach is the way we determine σ_{opt} . In the case of a known form of $\phi(r)$, the typical way to estimate the effective hard-sphere diameter is using the Barker-Henderson (BH) approach, namely [23],

$$\sigma_{\text{BH}} = \int_0^\infty \left[1 - e^{-\beta \phi^{\text{ref}}(r)} \right] dr, \quad (2)$$

where $\beta = (k_B T)^{-1}$ and ϕ^{ref} can be obtained following various schemes. For example, ϕ^{ref} can be defined using the Weeks-Chandler-Anderson (WCA) splitting [24], in which the potential is truncated at a distance corresponding to its minimum and shifted so that it vanishes at that point. However, here we assume that the pair potential, $\phi(r)$, is unknown and therefore adopt a different approach.

A. Effective hard-sphere diameter

The simplest way to estimate an effective diameter directly from a RDF is from the largest distance at which $g(r) = 0$, yielding σ_u . A better estimate can be obtained by first recovering an effective interaction potential from the RDF using the IBI method (see details in Sec. 1 of the Appendix). The effective IBI potential can be split using the WCA approach, providing an estimate for the reference potential, $\phi^{\text{ref}}(r)$. Then, the OZ equation, together with an appropriate closure relation, here the Percus-Yevick (PY) closure [25, 26], is solved to obtain the corresponding correlation function $g^{\text{ref}}(r)$. The optimal hard-sphere diameter σ_{opt} is determined by minimizing

$$\mathcal{D}(\sigma_{\text{opt}}) = \int_0^{r_c} [g^{\text{ref}}(r) - g^{\text{HS}}(r; \sigma_{\text{opt}})]^2 dr, \quad (3)$$

where $g^{\text{HS}}(r; \sigma_{\text{opt}})$ denotes the hard-sphere RDF corresponding to the effective diameter σ_{opt} . Noise is often present in a RDF measured either experimentally or from simulations. To minimize the influence of such noise, we introduce a cutoff distance, r_c , based on the positions of the peaks in $g(r)$. Specifically, we determine the average separation between successive peaks, denoted by r_p . For distances greater than $r_{\text{last}} + r_p$, where r_{last} is the position of the last identified peak, the measured RDF is set to unity beyond $r_c = r_{\text{last}} + r_p$, i.e., $g(r) = 1$ for $r > r_c$. In systems exhibiting only a single peak, which typically occurs at low densities, the cutoff distance is taken as $r_c = 2r_p$.

Multi-component systems, for which the same idea applies, are discussed in more detail in the Appendix; see also Refs. [19, 27].

For systems interacting purely through soft potentials, the excluded-volume contribution vanishes and no effective size estimation is required. In such cases, the thermodynamics can be computed directly from the RDF and interaction potential as detailed in the next subsection.

B. Helmholtz free energy

To extract the thermodynamics of a system purely from structural data, we begin by expressing the total Helmholtz free energy as [13, 14]

$$F[\rho(\mathbf{r})] = F^{\text{id}}[\rho(\mathbf{r})] + F^{\text{exc}}[\rho(\mathbf{r})], \quad (4)$$

where the first term represents the exact ideal-gas contribution, $\beta F^{\text{id}}[\rho] = \int_V d\mathbf{r} \rho(\mathbf{r}) [\ln(\Lambda^3 \rho(\mathbf{r})) - 1]$, where ρ is the one-body number density, Λ is the de Broglie wavelength, and V the volume of the system. $F^{\text{exc}}[\rho(\mathbf{r})]$ denotes the excess free energy functional describing particle interactions via the pair potential ϕ , which can be decomposed as in Eq. (1). In the same way, the excess free energy functional can be split into

$$F^{\text{exc}}[\rho(\mathbf{r})] = F_{\text{ref}}^{\text{exc}}[\rho(\mathbf{r})] + F_{\text{tail}}^{\text{exc}}[\rho(\mathbf{r})]. \quad (5)$$

Here, $F_{\text{ref}}^{\text{exc}}$ defines the reference-system free energy arising from excluded volume effects through ϕ^{ref} , while $F_{\text{tail}}^{\text{exc}}$ describes the long range attractive (or repulsive) interactions treated perturbatively through ϕ^{tail} . By mapping ϕ^{ref} onto an effective hard-sphere system as described above, $F_{\text{ref}}^{\text{exc}}$ can be approximated by the excess free energy of the corresponding hard-sphere fluid. For the bulk fluid, the PY approximation [20, 21] or even better, the Carnahan-Starling equation of state [22, 28] provide suitable approximations. Here, we develop the formalism to include inhomogeneous fluids, so $F_{\text{ref}}^{\text{exc}}$ may therefore be evaluated using fundamental measure theory (FMT), which in bulk generates either the PY or Carnahan-Starling equation of state, depending on the particular implementation [29, 30].

The remaining contribution to be determined, $F_{\text{tail}}^{\text{exc}}$, is described in the next section.

C. Excess free energy

A widely used approach for obtaining the free energy contribution from the tail is via thermodynamic integration [13]. The often used starting point is the result that the functional derivative of the free energy with respect to the pair potential $\phi(r)$ is related to the two-body density distribution function $\rho^{(2)}(\mathbf{r}, \mathbf{r}')$, as follows [14]

$$\rho^{(2)}(\mathbf{r}, \mathbf{r}') = 2 \frac{\delta F[\rho]}{\delta \phi(\mathbf{r}, \mathbf{r}')}.$$
 (6)

Starting from the free energy of a reference system with pair potential $\phi^{\text{ref}}(r)$ and ‘turning on’ the tail of the potential via the coupling parameter $\alpha \in [0, 1]$, so that $\phi_\alpha(r) = \phi^{\text{ref}}(r) + \alpha \phi^{\text{tail}}(r)$, yields a following formally exact result for $F_{\text{tail}}^{\text{exc}}[\rho(\mathbf{r})]$, namely [14, 31]

$$F_{\text{tail}}^{\text{exc}}[\rho(\mathbf{r})] = \frac{1}{2} \int_0^1 d\alpha \int d\mathbf{r} \int d\mathbf{r}' \rho^{(2)}(\mathbf{r}, \mathbf{r}'; \alpha) \phi^{\text{tail}}(|\mathbf{r} - \mathbf{r}'|),$$
 (7)

where $\rho^{(2)}(\mathbf{r}, \mathbf{r}'; \alpha)$ denotes the two-body density at coupling parameter α . Writing $\rho^{(2)}(\mathbf{r}, \mathbf{r}'; \alpha) = \rho(\mathbf{r})\rho(\mathbf{r}')g_\alpha(\mathbf{r}, \mathbf{r}')$ [13], where g_α is the RDF for the system at α , the attractive contribution becomes

$$F_{\text{tail}}^{\text{exc}}[\rho(\mathbf{r})] = \frac{1}{2} \int d\mathbf{r} \int d\mathbf{r}' \rho(\mathbf{r})\rho(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') \phi^{\text{tail}}(|\mathbf{r} - \mathbf{r}'|),$$
 (8)

where

$$G(\mathbf{r}, \mathbf{r}') = \int_0^1 d\alpha g_\alpha(\mathbf{r}, \mathbf{r}') \quad (9)$$

is the coupling-averaged pair correlation function.

For a homogeneous fluid, where $\rho(\mathbf{r}) = \rho_0$, we have $\rho^{(2)}(\mathbf{r}, \mathbf{r}'; \alpha) = \rho_0^2 g_\alpha(r)$, where $r = |\mathbf{r} - \mathbf{r}'|$, yielding

$$\frac{F_{\text{tail}}^{\text{exc}}[\rho_0]}{V} = \frac{F_{\text{ref}}^{\text{exc}}[\rho_0]}{V} + \frac{1}{2} \rho_0^2 \int_0^1 d\alpha \int d\mathbf{r} g_\alpha(r) \phi^{\text{tail}}(r),$$
 (10)

where the last term corresponds to $F_{\text{tail}}^{\text{exc}}[\rho_0]/V$.

We now define the integrated interaction strength

$$A_{\text{TI}} = 4\pi \int_0^\infty r^2 G(r) \phi^{\text{tail}}(r) dr = 4\pi \int_0^\infty r^2 a_{\text{TI}}(r) dr, \quad (11)$$

where $a_{\text{TI}}(r) = G(r) \phi^{\text{tail}}(r)$ is the integration kernel. Using Eq. (11), the excess free energy for the bulk system can be written as

$$\frac{F^{\text{exc}}[\rho_0]}{V} = \frac{F_{\text{ref}}^{\text{exc}}[\rho_0]}{V} + \frac{1}{2} \rho_0^2 A_{\text{TI}}. \quad (12)$$

Here, for simplicity, we approximate a_{TI} , and hence the corresponding integrated strength A_{TI} , using

$$a_{\text{TI}}(r) = g(r) \phi^{\text{tail}}(r), \quad (13)$$

where $\phi^{\text{tail}}(r)$ is the attractive tail obtained from the IBI procedure. This approximation assumes that $g_\alpha(r)$ remains approximately unchanged as the attractive tail is gradually introduced, such that $g_\alpha(r) \simeq g_{\alpha=1}(r)$ over the integration path. This is motivated by the assumption that the structural correlations are dominated by the excluded-volume interactions, while the attractive tail provides a comparatively weaker contribution to the structure. Consequently, the full-system RDF, $g_{\alpha=1}(r)$, can be used to approximate the structural correlations throughout the integration, eliminating the need to evaluate $g_\alpha(r)$ at multiple values of α .

D. Excess free energy via $c^{(2)}(r)$

Instead of starting from Eq. (6) and integrating to obtain the free energy in Eq. (7), an alternative approach is to start from the following result that the pair direct correlation function is the second derivative of the free energy as a function of the one-body density, namely [13, 14]:

$$c^{(2)}(\mathbf{r}, \mathbf{r}') = -\beta \frac{\delta^2 F[\rho]}{\delta \rho(\mathbf{r}) \delta \rho(\mathbf{r}')}. \quad (14)$$

Integrating this twice, we obtain the following for the excess free energy [14]

$$\beta F^{\text{exc}}[\rho(\mathbf{r})] = \int_0^1 d\alpha \times \int d\mathbf{r} \int d\mathbf{r}' \rho(\mathbf{r})\rho(\mathbf{r}') c_\alpha^{(2)}(\mathbf{r}, \mathbf{r}'), \quad (15)$$

where again $\alpha \in [0, 1]$, and the integration is through a series of states with density profile $\alpha\rho(\mathbf{r})$. The function $c_\alpha^{(2)}(\mathbf{r}, \mathbf{r}')$ is the corresponding DCF along the integration path. Splitting $c_\alpha^{(2)}$ into a part corresponding to a reference system (excluded volume contribution), $c_{\text{ref}}^{(2)}$, and a remainder (long range tail), this gives

$$F^{\text{exc}}[\rho(\mathbf{r})] = F_{\text{ref}}^{\text{exc}}[\rho(\mathbf{r})] + k_B T \int_0^1 d\alpha \int d\mathbf{r} \int d\mathbf{r}' \rho(\mathbf{r})\rho(\mathbf{r}') \Delta c_\alpha^{(2)}(\mathbf{r}, \mathbf{r}'), \quad (16)$$

where $\Delta c_\alpha^{(2)} = c_\alpha^{(2)} - c_{\text{ref}}^{(2)}$. In bulk this gives [15]

$$\frac{F^{\text{exc}}[\rho_0]}{V} = \frac{F_{\text{ref}}^{\text{exc}}[\rho_0]}{V} + k_B T \rho_0^2 \int_0^1 d\alpha \int d\mathbf{r} \Delta c_\alpha^{(2)}(r). \quad (17)$$

By assuming $\int_0^1 d\alpha \Delta c_\alpha^{(2)}(r) \approx -\frac{1}{2} \Delta c^{(2)}(r)$, we get

$$\frac{F^{\text{exc}}[\rho_0]}{V} = \frac{F_{\text{ref}}^{\text{exc}}[\rho_0]}{V} - \frac{1}{2} k_B T \rho_0^2 \int d\mathbf{r} \Delta c^{(2)}(r), \quad (18)$$

where in this approximation, we assume that the value of $\Delta c_\alpha^{(2)}(r)$ increases almost linearly from 0 to $\Delta c^{(2)}(r)$ by varying α from 0 to 1.

Comparing Eqs. (10) and (18), it is tempting to identify $G(r)\phi_{\text{tail}}(r)$ with $-k_B T \Delta c^{(2)}(r)$. However, this identification is not generally valid, as the two expressions arise from different approximations and therefore need not be equivalent. However, in analogy with Eq. (11), we do denote the integral in (18) as

$$A_{c^{(2)}} = -k_B T 4\pi \int r^2 \Delta c^{(2)}(r) dr, \quad (19)$$

where $A_{c^{(2)}}$ is the integrated attraction strength obtained via the DCF route. Here, the integrated attraction strength kernel is given as $a_{c^{(2)}} = -k_B T \Delta c^{(2)}(r)$.

Here, we compute $c_{\text{ref}}^{(2)}$ for the reference system via the OZ equation (see Appendix) together with the PY closure, where the reference system is approximated by an effective hard-sphere fluid with diameter σ_{opt} , following the procedure described in Sec. II A.

E. Polymer free energy

As mentioned at the beginning of this section, if $g(r) > 0 \forall r$, there is no reason to define $F_{\text{ref}}^{\text{exc}}$. This feature of the RDF can be seen in several models for, e.g., linear polymers and dendrimers [5, 32–40]. In such systems, in bulk, the total excess free energy reduces to [5]

$$F^{\text{exc}}[\rho_0] = \frac{1}{2} \rho_0^2 A V, \quad (20)$$

with integrated strength

$$A = 4\pi \int_0^\infty r^2 \phi(r) dr, \quad (21)$$

with $\phi(r)$ being the (unknown), soft, full interaction potential. This can be formally obtained by setting $G(r) = 1$ (which corresponds to neglecting correlations, though with smaller impact than one might expect, see [31]) in Eq. (8) and $\phi(r) = \phi^{\text{tail}}(r)$ in Eq. (1). Alternatively, assuming $c_{\text{ref}}^{(2)}(r) = 0$ in Eq. (18) provides the integrated strength

$$A_{c^{(2)}} = -k_B T 4\pi \int r^2 c^{(2)}(r) dr. \quad (22)$$

Comparing Eq. (21) with Eq. (22) recovers the random phase approximation (RPA) result, namely $\phi(r) = -k_B T c^{(2)}(r)$. For soft particles the RPA has been shown to provide satisfactory results [5, 32, 40–45].

Equations (18), (19) and (22) establish a direct link between the microscopic structure and the thermodynamics, since the DCF can be obtained from scattering experiments via the OZ relation [46–48], or from simulations and optical measurements through the radial distribution function $g(r)$, as demonstrated in [49].

F. Relation between second virial coefficient and integrated strength

Near the binodal of phase-separating system, accurately determining the DCF becomes increasingly difficult, and numerical solutions of the OZ equation with standard closures often fail to converge [18]. To address this, we employ an alternative approach based on the second virial coefficient to estimate the integrated interaction strength, which within the present formalism serves as an initial estimate of the coexistence properties.

The second virial coefficient is defined as [50]

$$B = -2\pi \int_0^\infty \left[e^{-\beta\phi(r)} - 1 \right] r^2 dr. \quad (23)$$

In the dilute limit, the RDF reduces to the Boltzmann factor [13], given as

$$g_\alpha(r) \simeq e^{-\beta\phi_\alpha(r)} = e^{-\beta[\phi^{\text{ref}}(r) + \alpha\phi^{\text{tail}}(r)]}. \quad (24)$$

This allows $G(r)$ to be evaluated analytically by integrating Eq. (24) with respect to α (as given in Eq. (9)), leading to

$$G(r) = e^{-\beta\phi^{\text{ref}}(r)} \frac{1 - e^{-\beta\phi^{\text{tail}}(r)}}{\beta\phi^{\text{tail}}(r)}. \quad (25)$$

Substituting the latter into Eq. (11) yields

$$G(r)\phi^{\text{tail}}(r) = \frac{e^{-\beta\phi^{\text{ref}}(r)} \left[1 - e^{-\beta\phi^{\text{tail}}(r)} \right]}{\beta}. \quad (26)$$

By integrating as in Eq. (11), one gets the integrated strength

$$\frac{4\pi}{\beta} \int_0^\infty r^2 \left(e^{-\beta\phi^{\text{ref}}(r)} - e^{-\beta\phi(r)} \right) dr = 2\beta^{-1} (B - B_{\text{ref}}), \quad (27)$$

where B_{ref} is the virial coefficient of the reference system. Combining Eqs. (26) and (27) with Eq. (11), we obtain

$$A_B = 2\beta^{-1} (B - B_{\text{ref}}) = 2\beta^{-1} \Delta B, \quad (28)$$

where A_B denotes the integrated attraction strength expressed in terms of the difference in second virial coefficients between the reference and full system. As one might expect, this virial-level description becomes inaccurate at intermediate and high densities due to the increasing role of many-body correlations.

G. Colloid-polymer mixture

The formalism developed for one-component systems can be readily extended to multi-component mixtures. Here, as an example, we consider an n -component mixture consisting of both colloidal and polymeric species. For all pair interactions that feature a hard-core repulsion (e.g. colloid-colloid and colloid-polymer interactions) the free energy is decomposed as in Eq. (5).

The reference system is chosen such that colloid-colloid and colloid-polymer interactions include excluded-volume effects, while polymer-polymer interactions are treated solely with the RPA. Accordingly, the excess free energy of the reference system is

$$F_{\text{exc}}^{\text{ref}}[\{\rho_i\}] = F_{\text{hs}}[\{\rho'_i\}] + F_d[\{\rho_i\}] + F_\phi[\{\rho_i\}], \quad (29)$$

where F_{hs} accounts for hard-sphere correlations (colloid-colloid), F_d represents depletion effects arising from non-additivity (colloid-polymer). Here, $\{\rho'_i\}$ denotes the subset of densities corresponding to colloidal species, while $\{\rho_i\}$ includes all components. $F_\phi[\{\rho_i\}]$ accounts for the contributions from the pairs interacting via purely soft potential such as polymer-polymer and has the same form as given by Eq. (20) and (22). As for simplicity, for such pairs i and j the reference free energy contribution is expressed as

$$F_\phi[\{\rho_i\}] = \frac{1}{2} \sum_{i,j} A_{ij} \rho_i \rho_j \quad (30)$$

where A_{ij} is given as identical to the one-component polymeric system,

$$A_{ij} = -k_B T \int d\mathbf{r} c_{ij}^{(2)}(r). \quad (31)$$

Within this framework, the full excess free energy can be expressed in a mean-field-like form as

$$F_{\text{exc}} = F_{\text{exc}}^{\text{ref}}[\{\rho_i\}] + \frac{1}{2} \sum_{i,j} A_{ij} \rho_i \rho_j, \quad (32)$$

where A_{ij} describes the integrated strength of the tails contribution between species i and j , interacting via excluded volume effect, while $F_{\text{exc}}^{\text{ref}}[\{\rho_i\}]$ is the reference system excess free energy as already given in Eq. (29). For any such pairs, wherever the decomposition is applicable (e.g., LJ potential, generalized Mie potential), the coefficient A_{ij} can be expressed in terms of the difference of DCF (as given by Eq. (19)),

$$A_{ij} = -k_B T \int d\mathbf{r} \Delta c_{ij}^{(2)}(r), \quad (33)$$

where $\Delta c_{ij}^{(2)}(r) = c_{ij}^{(2)}(r) - c_{ij}^{(2),\text{ref}}(r)$.

A detailed evaluation of $F_{\text{hs}}[\{\rho'_i\}]$, based on the Rosenfeld fundamental measure theory (FMT) functional, and of $F_d[\{\rho_i\}]$, within the free-volume theory framework, is discussed in the given reference [10, 29, 51, 52]

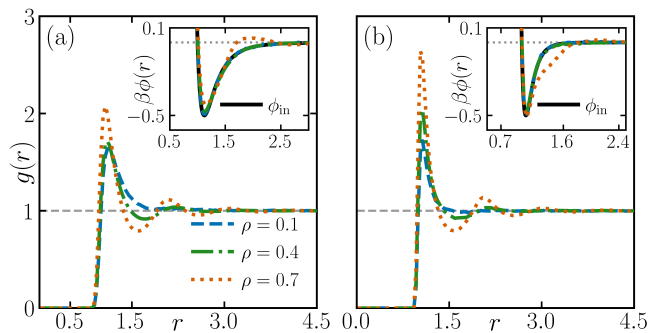


FIG. 1. Radial distribution functions, $g(r)$, from Langevin simulations for (a) 12–6 and (b) 24–12 interaction potentials with $\beta\epsilon = 0.5$. Results are displayed for the number densities $\sigma^3\rho = 0.1, 0.4$, and 0.7 . The insets show the corresponding pair potentials used in the simulations, ϕ^{real} (cyan), together with the reconstructed potentials obtained with the IBI from the corresponding RDFs.

III. RESULTS

A. One-component system

To validate our approach, we first consider a one-component colloidal fluid interacting through a generalized Mie [53, 54] potential of the form

$$\beta\phi^{\text{real}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^m - \left(\frac{\sigma}{r} \right)^n \right]. \quad (34)$$

Here, ϵ controls the attraction strength, while σ sets the size of the particles. In the following, we set $\sigma = 1$. We select two types of interaction. In the first case, $(m, n) = (12, 6)$, which corresponds to the standard (12–6) Lennard–Jones (LJ) potential, while in the second case, $(m, n) = (24, 12)$, which corresponds to a steeper potential with shorter range of attraction. For both cases, we generate RDFs by means of Langevin dynamics [55, 56].

Figure 1 shows the RDFs obtained from simulation and used as structural input for the 12–6 potential (panel (a)) and for the 24–12 potential (panel (b)). These results are for $\beta\epsilon = 0.5$, i.e., corresponding to a temperature above the critical region of the corresponding fluid. The insets compare the original pair potentials used in the simulations with the potentials reconstructed from the RDFs using the IBI procedure, where the details of the procedure are provided in the Supplementary Material. At low and moderate densities, the reconstructed potentials are in good agreement with the underlying interaction potentials. At higher densities, however, the convergence of the inversion becomes slower, and the reconstructed potentials may deviate from the original ones. This does not compromise the present approach, since the IBI potential serves only as an intermediate quantity for identifying the repulsive reference contribution and estimating the effective hard-sphere diameter entering the reference

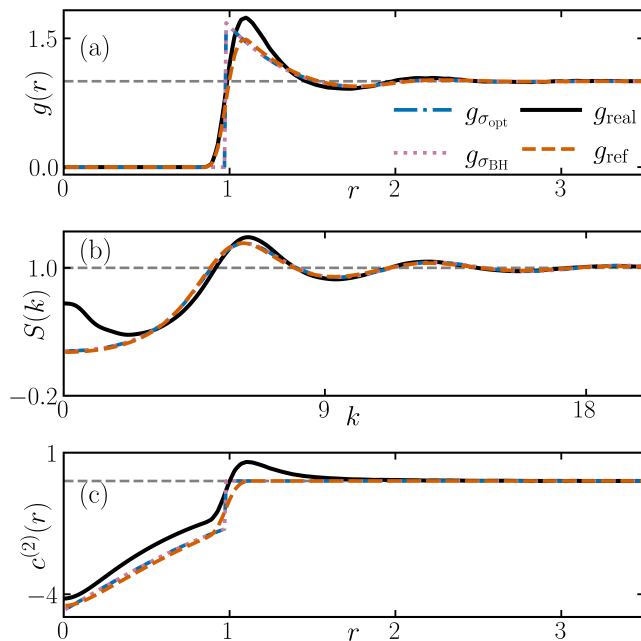


FIG. 2. Structural correlation functions for a one-component fluid interacting through a 12-6 LJ potential with $\beta\epsilon = 0.5$ and $r_{\text{cutoff}} = 3.5$, at number density $\sigma^3\rho = 0.4$. The black solid curves represent the real system, with the input RDF, $g(r)$, obtained from Langevin dynamics simulations. Panel (a) shows the radial distribution function, $g(r)$; panel (b) the static structure factor, $S(k)$; and panel (c) the direct correlation function, $c^{(2)}(r)$. The vermilion dashed curves represent the reference system obtained by retaining only the repulsive contribution after WCA decomposition of the IBI-reconstructed interaction potential. The blue long-dashed curves correspond to an effective hard-sphere reference system with the diameter determined by RDF optimization [see Eq. (3)], whereas the pink short-dashed curves represent the hard-sphere reference system with the diameter determined using the Barker-Henderson (BH) prescription.

free energy at each density.

We next examine how the structural information is mapped onto the reference system to determine the effective particle diameter. In experiments, the static structure factor, $S(k)$, is often directly accessible, while the corresponding radial distribution function, $g(r)$, can be obtained through the standard Fourier-transform relation between $S(k)$ and $g(r)$ [13]. In simulations, the radial distribution function is readily computed from particle configurations, from which the static structure factor can subsequently be obtained. To get the reference system, first, we decompose the inverted potential using the WCA scheme, allowing us to isolate the repulsive reference contribution, which is then mapped onto an effective hard-sphere system through the diameter-fitting procedures discussed in Sec. II A. Figure 2 compares the structure factor, RDF, and DCF of the real system with those of the corresponding reference systems. The latter include the WCA reference system, constructed from

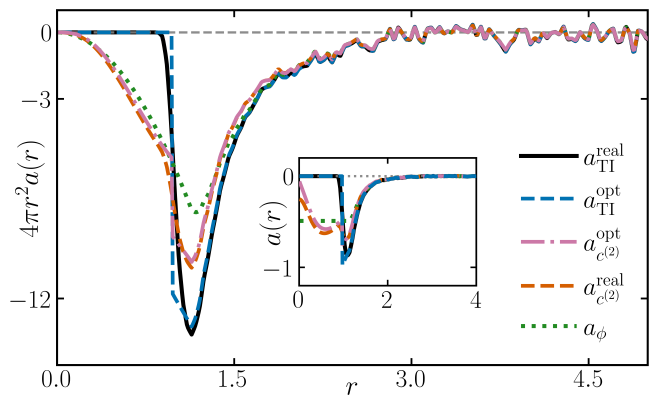


FIG. 3. Kernel $a(r)$ defined in (11) and (19) for a one-component system, computed using different routes. The green curve, a_ϕ , represents the attractive tail obtained by splitting the bare IBI potential [cf. Eq. (21)] for a prescribed $g(r)$. The RDF was computed using Langevin dynamics for a system of particles interacting via a 12-6 Lennard-Jones potential with $\beta\epsilon = 0.5$, $r_{\text{cutoff}} = 3.5$, and number density $\sigma^3\rho = 0.4$. The vermilion curve corresponds to the WCA-based repulsive reference, while the pink long-dashed curve corresponds to the optimized hard-sphere reference using $\Delta c^{(2)}$ route. The black solid curve shows the kernel obtained from the thermodynamic-integration route using the WCA reference, and the blue curve shows the corresponding result when the optimized hard-sphere reference is used.

the repulsive part of the inverted IBI potential, together with two hard-sphere approximations based on either BH diameter [Eq. (2)] or the optimized diameter, σ_{opt} , obtained by fitting the RDF using Eq. (3). The corresponding reference DCFs are computed by solving the OZ equation with the PY closure. The main source of deviation between the curves corresponding to the fully interacting system and the different reference systems is the absence of the contribution coming from the attractive tail in the latter case. The close agreement between the WCA, BH, and optimized hard-sphere references demonstrates that the excluded-volume effect is largely independent of the particular reference construction. Consequently, the excess free energy can be separated into a reference contribution, $F_{\text{ref}}^{\text{exc}}$, describing the excluded-volume interactions, and a tail contribution, $F_{\text{tail}}^{\text{exc}}$, describing the attractive interactions.

$F_{\text{ref}}^{\text{exc}}$ can be evaluated using, e.g., the White Bear version of FMT [28, 30, 52], while $F_{\text{tail}}^{\text{exc}}$ depends on the integrated interaction strength A , which is obtained by integrating the kernel $a(r)$; see Eqs. (11) and (19). In Fig. 3 we compare the kernels obtained using the different routes discussed in Sec. II. The kernels obtained from the DCF route, using either the WCA reference ($a_{c^{(2)}}^{\text{real}}$) or the optimized hard-sphere reference ($a_{c^{(2)}}^{\text{opt}}$), remain close to those obtained from the thermodynamic-integration route ($a_{\text{TI}}^{\text{real}}$) with WCA reference and with σ_{opt} reference ($a_{\text{TI}}^{\text{opt}}$). This agreement shows that the attractive contribution extracted from the structural correlations

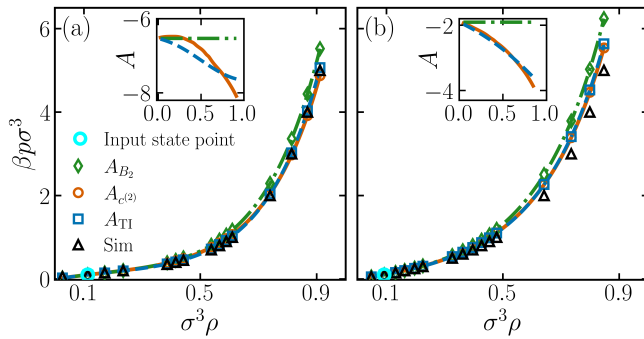


FIG. 4. Pressure equation of state for a one-component fluid interacting through (a) the Lennard-Jones 12-6 potential and (b) the generalized Mie 24-12 potential. In both cases $\beta\epsilon = 0.5$. Black triangles denote simulation results. Green diamonds show pressures obtained using the integrated strength derived from the virial route, blue squares correspond to the TI route, and vermillion circles denote the DCF route. The inset shows the corresponding integrated strength as a function of density. The open cyan circle marks the single simulation state point used as structural input for constructing the reference system and evaluating the density-dependent integrated strength.

is robust with respect to the precise representation of the repulsive reference system. In contrast, the kernel associated directly with the attractive part of the WCA-decomposed potential differs more substantially.

To assess the accuracy of the free energies obtained using the different routes, Figs. 4(a) and (b) present the corresponding equations of state together with direct simulation results for the 12-6 and 24-12 interaction potentials, respectively. The only structural input required here is obtained from the single state point indicated by the open circle. From this state point, the RDF is used to reconstruct the reference description, estimate the effective particle diameter, and evaluate the integrated strength, and the pressure is then predicted over a range of densities.

For both cases, the TI and the DCF routes show significantly better agreement with simulation than the virial route. This is expected, as the virial route is rooted in the dilute-limit description and therefore cannot fully capture the density dependence of the effective attractive contribution. This limitation is visible in the inset of Fig. 4, where the integrated strength obtained from the virial is independent on the density. In contrast, the TI and DCF routes incorporate structural information beyond the dilute limit and therefore provide a more reliable equation of state at moderate and high densities.

We also test the consistency of our approach by comparing the binodal of the cut-and-shifted LJ fluid, with $r_{\text{cutoff}} = 3.5$ obtained in Ref. [57], with the binodal obtained from the TI, DCF and virial routes. The results are showed in Fig. 5. Overall, we observe good agreement, particularly in the prediction of the gas branch of the binodal. The critical point is slightly underesti-

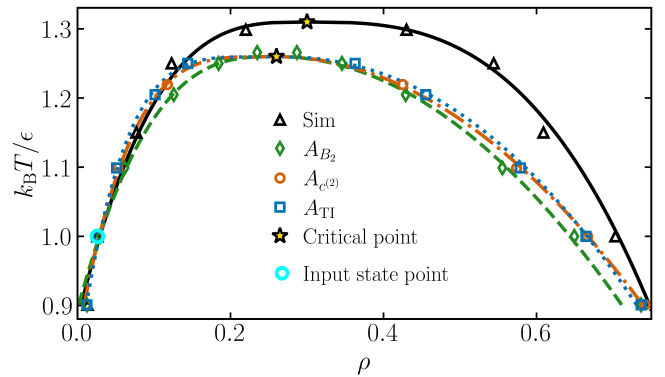


FIG. 5. Binodal curves for a LJ fluid obtained using the different free energy routes. The simulation data (Sim) is obtained from Ref. [57]. Star symbols denote the corresponding critical points. The cyan open circle marks the single state point, $(\sigma^3\rho, k_B T\epsilon^{-1}) = (0.05, 1)$, used as structural input.

mated, as is most of the liquid branch. Some of these deviations can be attributed to the truncation and shifting of the interaction potential in the simulations. Nevertheless, all three routes reproduce the overall shape of the coexistence curve, with the TI route showing the best agreement, followed by the DCF route and, finally, the virial route.

It is also important to note that the critical points obtained from the different integral-equation-based routes are practically overlapping because the initial coexistence estimate is generated from the virial route. After this initial step, the integrated strength is updated using either the TI or DCF route, and the coexistence densities are recomputed iteratively until convergence.

Taken together, the results so far demonstrate that accurate thermodynamic properties and phase behaviour can be recovered directly from structural correlations without requiring explicit interaction potentials. Moreover, near the critical point and within the coexistence region, where homogeneous structural data become increasingly difficult to obtain, the smooth density dependence of the integrated strength A permits reliable interpolation or extrapolation into these regions, further extending the applicability of the proposed framework.

B. Two-component colloid-polymer mixture

We now extend our approach to a binary colloid-polymer mixture. This system provides a more demanding test than the one-component fluid, because the thermodynamics is controlled not only by the correlations within each species, but also by the cross correlations between colloids and polymers. The aim is to determine whether structural information obtained at a single state point contains enough information to construct the integrated interaction strengths, A_{ij} , and hence the equation of state and the coexistence phase behavior of the mix-

ture.

The polymer-polymer interaction used to generate the reference input data is given by the Gaussian potential [33, 36],

$$\beta\phi_{pp}(r) = \epsilon_{pp} \exp\left(-\frac{r^2}{R_{pp}^2}\right), \quad (35)$$

with $\beta\epsilon_{pp} = 2.0$ and $R_{pp} = \sigma = 1$. The colloid-polymer interaction is kept purely repulsive and is represented by a Weeks-Chandler-Andersen (WCA) [24] potential with $\beta\epsilon_{cp} = 1.0$, $\sigma_{cp} = 1.0$, and $r_{\text{cutoff}} = 1.1224$. For the colloid-colloid interaction, we consider generalized Mie potentials with exponent pairs 12-6. The parameters are fixed at $\beta\epsilon_{cc} = 0.5$, $\sigma_{cc} = 1.0$, and $r_{\text{cutoff}} = 3.5$, so that the attractive part of the colloid-colloid interaction is retained.

The structural correlations of the full system are obtained from Langevin dynamics simulations at polymer and colloid number densities $\sigma^3\rho_p = 0.6$ and $\sigma^3\rho_c = 0.09$, respectively. The corresponding radial distribution functions are shown as black solid curves in Fig. 6(a)–(c) for the polymer-polymer, colloid-polymer, and colloid-colloid pairs, respectively. As in the one-component case, the IBI procedure is used only as an intermediate step to construct a state-point-specific decomposition of the effective interactions and to determine the corresponding excluded-volume reference interactions.

Following the construction of the effective pair potentials, we define the corresponding reference systems for those pair interactions involving excluded-volume effects, namely the colloid-polymer and colloid-colloid interactions. The repulsive reference interactions are constructed using the WCA decomposition and, alternatively, by mapping the interactions onto effective hard-sphere systems with diameters obtained from the optimized RDF fit, σ_{opt} , and from the Barker-Henderson prescription, σ_{BH} . The resulting RDFs for the different reference descriptions are compared with those of the full system in Fig. 6(a)–(c).

For the polymer-polymer pair [Fig. 6(a)], the interaction is purely soft and is retained explicitly in the reference system description. Consequently, the reference and full-system RDFs coincide. The colloid-polymer interaction [Fig. 6(b)] is purely repulsive and therefore contains no attractive tail beyond the excluded-volume reference contribution. Accordingly, only minor differences are observed between the full and reference-system RDFs. In contrast, the colloid-colloid RDF [Fig. 6(c)] changes more noticeably after removal of the attractive contribution. Nevertheless, the positions of the principal peaks remain nearly unchanged, indicating that the attraction predominantly modifies the strength of the correlations rather than their characteristic structural length scale.

The weak response of the polymer-polymer correlations in Fig. 6(a) to the modification of the colloid-colloid interaction further indicates that the polymer distribution is only weakly affected by the attractive colloidal contribution. This behavior is consistent with the

reservoir-like role of the polymer component underlying the free-volume description [51].

The influence of the reference decomposition becomes more apparent in reciprocal space. The polymer-polymer, colloid-polymer, and colloid-colloid static structure factors are shown in Fig. 6(d), Fig. 6(e), and Fig. 6(f), respectively. While $S_{pp}(k)$ [Fig. 6(d)] remains comparatively insensitive to the choice of reference system, more pronounced differences are observed for $S_{cp}(k)$ [Fig. 6(e)] and $S_{cc}(k)$ [Fig. 6(f)], particularly in the small- k regime. The enhanced sensitivity of these correlations at long wavelengths reflects the role of the colloid-colloid attraction in controlling collective density fluctuations and, consequently, the phase behavior of the mixture.

A similar distinction is observed in the direct correlation functions shown in Fig. 6(g)–(i). The polymer-polymer DCF, $c_{pp}^{(2)}(r)$ [Fig. 6(g)], remains unchanged because the full soft polymer interaction is retained in the reference description. Likewise, only small differences are observed for the colloid-polymer DCF, $c_{cp}^{(2)}(r)$ [Fig. 6(h)], consistent with the absence of an attractive tail in this interaction. In contrast, $c_{cc}^{(2)}(r)$ [Fig. 6(i)] exhibits a clear deviation between the full and reference systems. This difference isolates the non-reference contribution associated with the attractive colloid-colloid interaction and enters directly into the integrated-strength correction discussed below.

Once the effective hard-sphere diameters, $\sigma_{\text{opt},cc}$ and $\sigma_{\text{opt},cp}$, have been determined, the excess free energy of the corresponding reference mixture can be constructed using Eq. (29). For the present nonadditive colloid-polymer mixture, the hard-sphere contribution, given by the first term of Eq. (29), is evaluated using the White Bear Mark I fundamental-measure functional for hard-sphere mixtures [30]. The colloid-polymer depletion contribution, represented by the second term, is described using the multicomponent free-volume theory [51]. The effective polymer size entering this contribution is determined from the additive relation between the characteristic length scales of the colloid-colloid and colloid-polymer interactions.

The remaining contribution to the reference free energy accounts for the soft polymer-polymer interaction. Since this pair interaction does not require a hard-core reference, no reference DCF subtraction is introduced. Instead, its thermodynamic contribution is evaluated directly from the full soft interaction using the DCF or virial route, as described in Sec. II G and Eq. (22). Thus, the reference free energy contains the hard-core colloidal contribution, the colloid-polymer depletion contribution, and the explicitly treated soft polymer-polymer interaction.

Having determined the reference free energy, the total excess free energy of the full interacting mixture is obtained from Eq. (32). The remaining quantities are the integrated interaction strengths, A_{ij} , associated with pair interactions that contain an excluded-volume reference. These quantities enter as coefficients of the

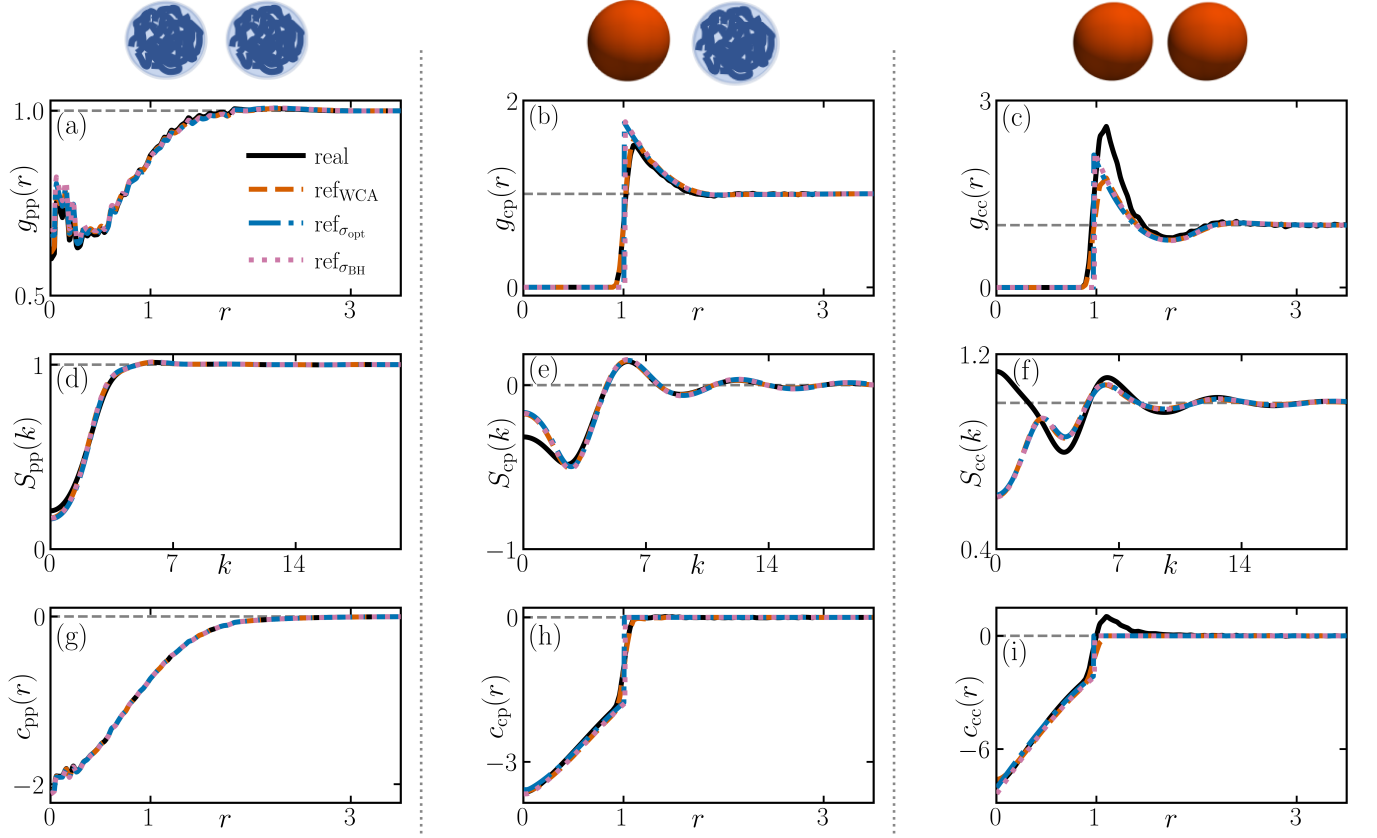


FIG. 6. Structural and correlation properties of a two-component colloid-polymer system. The left, middle, and right columns correspond to polymer-polymer, colloid-polymer, and colloid-colloid correlations, respectively. The first row shows the radial distribution functions $g_{ij}(r)$, the second row shows the static structure factors $S_{ij}(k)$, and the third row shows the direct correlation functions $c_{ij}^{(2)}(r)$. The black solid curves represent the full system obtained from simulation and serve as the structural input. The vermilion dashed curves denote the WCA reference system, obtained by retaining only the repulsive contributions. The blue long-dashed curves correspond to the effective hard-sphere reference system with diameter determined by RDF optimization, while the pink short-dashed curves represent the hard-sphere reference system with BH diameter. The colloid-colloid interaction is modeled by a LJ potential with $\epsilon_{cc} = 0.5$, $\sigma_{cc} = 1$, and $r_{\text{cutoff}} = 3.5$. The polymer-polymer interaction is modeled by a Gaussian potential with $\epsilon_{pp} = 2$ and $R_{pp} = 1.0$. The colloid-polymer interaction is represented by a purely repulsive LJ potential with $\epsilon_{cp} = 1.0$, $\sigma_{cp} = 1.0$, and $r_{\text{cutoff}} = 1.1224$. The densities are fixed at $\rho_p = 0.6$ and $\rho_c = 0.09$.

quadratic density contribution in the second term of Eq. (32). For such pairs, A_{ij} is obtained from the difference between the DCFs of the full and reference systems according to Eq. (33). Alternatively, A_{ij} may be evaluated using the virial or thermodynamic-integration routes, following the procedures introduced earlier for the one-component system.

In the present mixture, the only nonzero attractive perturbative term associated with an excluded-volume reference arises from the colloid-colloid interaction. The colloid-polymer pair, though interacting via excluded volume effect, contains no attractive tail beyond the reference contribution, and therefore $A_{cp} = 0$. Figure 7 shows the corresponding kernel, $a_{cc}(r)$, whose spatial integral yields the colloid-colloid integrated strength, A_{cc} . Results obtained from the thermodynamic-integration and DCF routes are compared and show good agreement. Likewise, the kernel obtained using the optimized hard-

sphere reference closely follows that obtained from the WCA reference, indicating that A_{cc} is only weakly sensitive to the specific representation of the repulsive core. The hard-core contribution to the reference free energy, however, remains strongly dependent on the effective particle diameter [58].

We next use the same structural input to evaluate the free energy and, subsequently, to predict the equation of state of the mixture. For this purpose, we apply the approach to two systems in which the colloid-colloid interaction is described by Mie potentials with exponents 12-6 and 24-12, while the polymer-polymer and colloid-polymer interactions are kept unchanged from those introduced earlier and specified in the caption of Fig. 8. The pressure is calculated as a function of the total density, ρ_t , at a fixed colloid fraction, $x = 0.16$. The structural input, $g(r)$, is obtained from Langevin dynamics simulations at the state point and for the interaction pa-

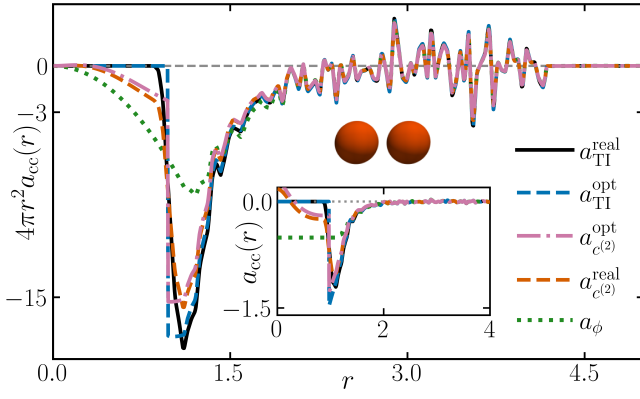


FIG. 7. Integrated-strength kernel $a(r)$ for the colloid-polymer interaction, computed using different routes. a_ϕ is obtained from the WCA splitting of the inverted IBI potential and is shown only as a reference for the inversion step. The thermodynamic contribution is evaluated through the TI and DCF routes. For the present parameter set, the contribution from the polymer-polymer pair are included in the reference free energy and the perturbative contribution from the colloid-polymer interaction remains negligible due to absence of the attractive tail.

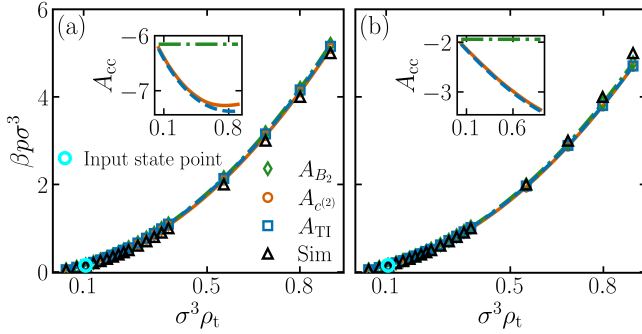


FIG. 8. Equation of state for the colloid-polymer mixture, shown as pressure versus total density ρ_t at fixed colloid fraction ($x=0.16$). The pressure is computed from the integrated strength obtained via the different routes indicated in the legend. Panels (a) and (b) correspond to colloid-colloid Mie interactions with exponent pairs 12-6 and 24-12, respectively. In both cases, $\epsilon_{cp} = 0.5$, $\sigma_{cp} = 1$, and $r_{\text{cutoff}} = 3.5$. The polymer-polymer interaction is Gaussian with $\epsilon_{pp} = 2$ and $R_{pp}^g = 1.0$, while the colloid-polymer interaction is purely repulsive with $\epsilon_{cc} = 1.0$, $\sigma_{cc} = 1.0$, and $r_{\text{cutoff}} = 1.1224$. The cyan circle marks the single simulation state point used as structural input.

rameters specified in the caption of Fig. 8. The cyan circle marks this single input state point. From the corresponding structural correlations, the matrix of integrated interaction strengths, A_{ij} , is evaluated and used to construct the bulk free energy over the full density range. The resulting free-energy description is then used to predict the pressure without requiring structural data at each individual density. For both the 12-6 and 24-

12 colloid-polymer interactions, the TI and DCF routes show very good agreement with the simulation data. The ΔB_2 route also performs better here than in the pure one-component attractive fluid. This improvement is physically reasonable because the colloid fraction is small, and the perturbative attractive contribution is therefore weighted less strongly in the total pressure. Nevertheless, the TI and DCF routes remain more reliable because they retain the structural information beyond the dilute-limit virial approximation. This is also reflected in the inset of Fig. 8, where the integrated strength varies only weakly with density for the mixture, making the structural-correlation routes particularly robust.

Finally, we use this approach to predict the fluid-fluid coexistence of a binary colloid-polymer mixture. This highlights the value of our overall approach: starting from structural correlations at a single state point, we construct the effective free energy and use it to determine bulk phase coexistence. The binodal is obtained by solving the coexistence conditions [13]. The chemical potentials and pressure are calculated from the free-energy, which includes the reference contribution and the integrated strengths A_{ij} .

For the binodal calculation, the polymer-polymer and colloid-polymer interactions are modeled as above, while for the colloid-colloid interaction we use a 24-12 Mie potential with $\sigma_{cc} = 1.0$, $r_{\text{cutoff}} = 2.5$, and $\beta\epsilon = 0.1$.

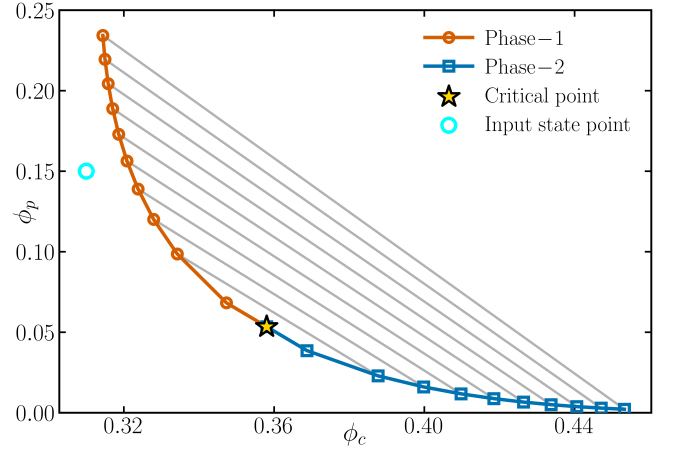


FIG. 9. Binodal curve for a binary colloid-polymer mixture predicted from structural correlations obtained at the input state point indicated. The integrated strengths entering the free energy are computed using the $\Delta c^{(2)}$ route. The two branches denote the coexisting polymer-rich/colloid-poor and colloid-rich/polymer-poor fluid phases.

The resulting binodal is shown in Fig. 9. The structural correlations at the input state point are used to compute the $\Delta c^{(2)}$ -based integrated strengths, which are then inserted into the binary free energy to predict the coexistence curve. The two branches of the binodal correspond to coexisting phases with different polymer and colloid concentrations. The appearance of a fluid-fluid binodal shows that the effective interactions encoded in

the structural correlations are sufficient to drive demixing, even when the polymers are nonideal and interact through a soft repulsive potential.

This result is important because most classical descriptions of colloid-polymer demixing rely on ideal-polymer assumptions, where the polymer fugacity can be related more directly to the polymer concentration. Once polymer-polymer interactions are introduced, this simplification is lost. The present formalism avoids this difficulty by using the structural correlations themselves to construct the free energy. In this sense, the binodal in Fig. 9 demonstrates that the same structural route used for the EOS can also be used to predict phase coexistence in an interacting colloid-polymer mixture.

The calculated binodal should nevertheless be interpreted as a fluid-fluid coexistence prediction within the homogeneous-fluid regime. At sufficiently high colloid density, a fluid-solid transition may become relevant, and this lies outside the scope of the present fluid-only formulation. Even with this limitation, the result provides a useful prediction of how soft polymer-polymer interactions, colloidal excluded volume, and cross correlations combine to control the stability of the mixed phase.

IV. CONCLUSION

In this work, we have developed a structural-correlation-based route for predicting the thermodynamics and phase behavior of colloidal and colloid-polymer mixtures. The main idea is to construct the free energy directly from radial distribution functions, static structure factors, and direct correlation functions, rather than relying on model interaction potentials. This is particularly useful for soft-matter systems of which the effective interactions are generally unknown experimentally, density dependent, and where many-body effects are important.

The formalism separates excluded-volume contribution from the one arising from long-range (attractive or repulsive) interactions. Interactions with a short-range repulsive core are decomposed into an effective hard-sphere reference contribution and a residual contribution, the latter being described by an integrated interaction strength, A . This quantity is evaluated using different routes, including thermodynamic integration, the direct correlation function route, and the virial route. In this framework, inverse Boltzmann iteration is used only as an intermediate step to construct the reference decomposition and estimate its effective contribution; the thermodynamic predictions are obtained from the integrated-strength formalism.

Our approach was first tested on two one-component colloidal systems. Using structural information from a single state point, we predicted the equation of state over a wide range of densities, in good agreement with simulated data. More specifically, the thermodynamic integration and direct correlation function routes showed

significantly better agreement with simulations than the virial route, especially at moderate and high densities, where dilute-limit approximation used within this route becomes inaccurate. Similarly, we have also predicted the binodal of one of the colloidal systems, proving that our approach can capture quantitatively the overall coexistence behavior from limited structural input.

We then extended our approach to binary colloid-polymer mixtures, where both self-correlations and cross-correlations contribute to the overall thermodynamics. From structural correlations obtained at a single input state point, we constructed the integrated-strength matrix, A_{ij} , and predicted the equation of state for a fixed composition. We found a quantitative agreement with simulated data, showing that our approach provides an effective means for describing system composed of both polymer and colloidal species.

Finally, we apply the framework to predict the fluid-fluid coexistence of a different colloid-polymer mixture. No comparison with simulation is attempted here; instead, the calculation serves to demonstrate the predictive power of our approach. Because the free-energy functional is constructed directly from structural correlations, without requiring explicit interaction potentials or a system-specific thermodynamic model, the approach is, in principle, transferable to arbitrary soft-matter systems for which structural information is available. The resulting binodal illustrates this capability.

All in all, the present framework provides a general route for constructing free-energy functionals directly from structural correlations, without requiring explicit knowledge of the underlying interaction potentials. This makes it applicable to a broad class of classical fluids for which structural information is available. Its current formulation nevertheless has some limitations. In particular, it has been developed for isotropic fluids and therefore does not directly account for orientational degrees of freedom arising in anisotropic particles or molecular fluids. Furthermore, for strongly correlated systems, such as highly charged colloidal or ionic suspensions, pair correlations alone may not provide a sufficiently accurate description of the excess free energy. In such cases, extensions incorporating more sophisticated reference functionals or higher-order correlation effects may be required. These directions provide natural avenues for future work.

ACKNOWLEDGMENTS

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APPENDIX

1. Iterative Boltzmann inversion

When the particle size is not known a priori, integral equation theory provides a systematic route to infer it from the radial distribution function. As an initial approximation, the interaction potential can be estimated in the low-density limit via $\beta\phi(r) = -\ln g(r)$. However, this relation breaks down at finite densities due to many-body correlations.

To overcome this limitation, we employ the iterative Boltzmann inversion (IBI) scheme,

$$\beta\phi^{(n)}(r) = \beta\phi^{(n-1)}(r) + \gamma \ln \left(\frac{g^{(n-1)}(r)}{g^{\text{target}}(r)} \right), \quad (36)$$

where γ is a mixing parameter controlling convergence and $g^{\text{target}}(r)$ denotes the target radial distribution function, obtained either from scattering experiments or from molecular simulations.

Starting from an initial guess for $\beta\phi^{(n)}(r)$, which is $-\ln(g(r))$ the corresponding radial distribution function is computed by solving the OZ equation (44) together with an appropriate closure relation. In this work, we used PY closure for the hard core interaction e.g., colloid-colloid and polymer-colloid and HNC for the soft interactions e.g., polymer-polymer.

Solving the OZ equation with these closures yields $h(r)$ and thus $g(r)$ for a given trial potential. The interaction potential is then updated via Eq. (36), and the procedure is iterated until convergence is achieved.

In the present implementation, this IBI framework is applied to a two-component system (colloids and polymers), where molecular dynamics simulations are first performed using the interaction potentials defined in the input model (Gaussian polymer-polymer interaction, WCA-like colloid-colloid repulsion, and Mie-type colloid-polymer interaction). The resulting simulation-generated radial distribution functions are then used as $g^{\text{target}}(r)$.

Starting from a physically motivated initial guess for the pair potentials, the IBI iterations reconstruct the effective interaction potentials that reproduce the simulated structure. As shown in Fig. 10, this inversion procedure successfully recovers the implemented interaction potentials for all three interactions (colloid-colloid, polymer-polymer and colloid-polymer), demonstrating the application of integral equation theory to recover the implemented potential.

In our IBI strategy for a multi-component system, we first supplied the one-component radial distribution function, $g(r)$, for each species in the mixture and treated each species as an independent one-component system to determine the corresponding self-interaction potentials. These potentials were then used as fixed inputs in the subsequent IBI calculations to determine the inter-species interaction potentials.

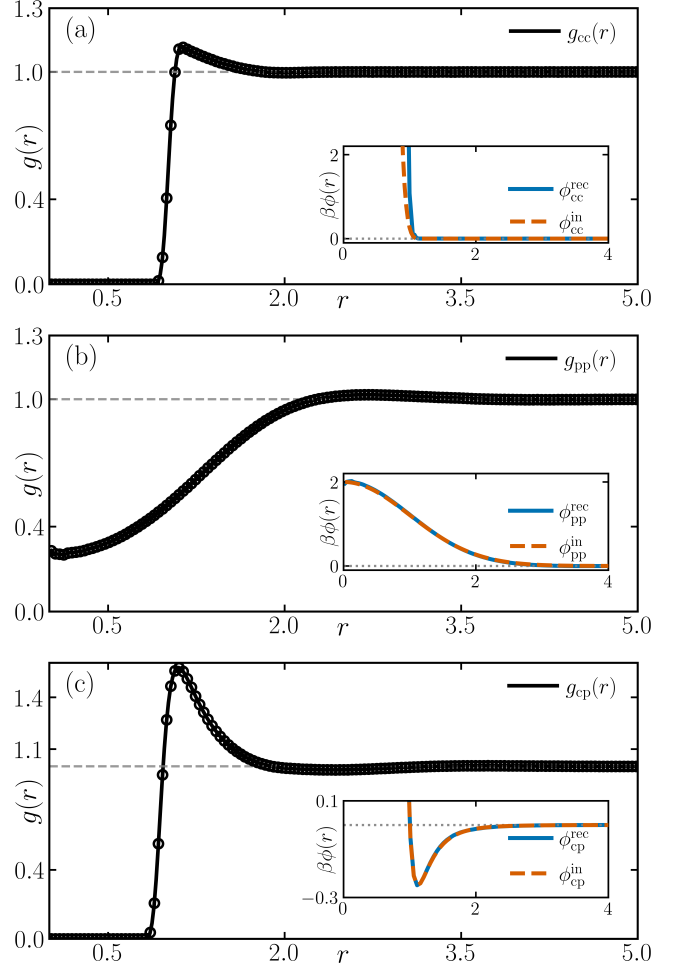


FIG. 10. $g(r)$ is obtained from the simulation is shown for the colloid-colloid (a), polymer-polymer (b) and the colloid-polymer (c) interactions. The corresponding applied potential in the simulation (dashed line) and the recovered potential (solid line) is shown in the inset, where the potential is recovered using IBI, at colloids densities $\rho_c = 0.01$ and $\rho_p = 0.54$.

Once the effective potential $\phi(r)$ is obtained, it is decomposed using the WCA splitting approach as

$$\phi(r) = \phi^{\text{ref}}(r) + \phi^{\text{tail}}(r) \quad (37)$$

such that

$$\phi^{\text{ref}}(r) = \begin{cases} \phi(r) - \phi(R), & r < R, \\ 0, & r \geq R, \end{cases} \quad (38)$$

while the attractive contribution is given by

$$\phi^p(r) = \begin{cases} \phi(R), & r < R, \\ \phi(r), & r \geq R, \end{cases} \quad (39)$$

where R denotes the position of the minimum of the potential, obtained from the correlation functions.

The reference potential $\phi^{\text{ref}}(r)$ isolates the repulsive core responsible for excluded volume effect. A corresponding reference radial distribution function $g^{\text{ref}}(r)$ is then computed and used as the new target. The effective particle size σ is determined by fitting a hard-sphere system whose radial distribution function best reproduces $g^{\text{ref}}(r)$.

For this purpose, the hard-sphere interaction is defined as

$$\phi_{\text{hs}}(r) = \begin{cases} \infty, & r < \sigma, \\ 0 & r \geq \sigma, \end{cases} \quad (40)$$

and the corresponding structure is obtained using the PY closure,

$$c_{\text{hs}}^{(2)}(r) = \begin{cases} -1, & r < \sigma_{ij}, \\ 0, & r \geq \sigma_{ij}, \end{cases} \quad (41)$$

It should be emphasized that the reconstruction of $\phi(r)$ is not an end in itself, but rather a means to determine the effective particle size and the associated direct correlation function. This step is crucial for formulating the free energy functional entirely in terms of structural quantities, thereby enabling a consistent connection between experimentally accessible correlations and thermodynamic properties, as developed in the following section.

2. Integral equation theory for a one component system

The structure of a homogeneous fluid is characterized by the radial distribution function $g(r)$, with the corresponding total correlation function defined as

$$h(r) = g(r) - 1. \quad (42)$$

Experimentally, structural information is typically obtained through the static structure factor $S(k)$, which is related to the Fourier transform of the total correlation function via

$$S(k) = 1 + \rho \hat{h}(k), \quad (43)$$

where ρ is the number density.

A formal connection between $h(r)$ and $c^{(2)}(r)$ is provided by the OZ equation,

$$h(r) = c^{(2)}(r) + \rho \int c^{(2)}(|\mathbf{r} - \mathbf{r}'|) h(r') d\mathbf{r}', \quad (44)$$

which, in Fourier space, reduces to

$$\hat{h}(k) = \frac{\hat{c}(k)}{1 - \rho \hat{c}(k)}. \quad (45)$$

This leads directly to

$$S(k) = \frac{1}{1 - \rho \hat{c}(k)}, \quad (46)$$

thereby establishing a direct link between experimentally measurable structural quantities and the direct correlation function. This relation forms the basis for expressing thermodynamic properties in terms of structural correlations.

3. Integral equation theory for a multi-component system

In a multicomponent fluid, the structure is characterized by the set of partial radial distribution functions $g_{ij}(r)$, with corresponding total correlation functions defined as

$$h_{ij}(r) = g_{ij}(r) - 1, \quad (47)$$

where the indices $i, j = 1, \dots, n$ denote correlations between species i and j in an n component system.

Experimentally accessible partial structure factors are related to $h_{ij}(r)$ via

$$S_{ij}(k) = \delta_{ij} + \sqrt{\rho_i \rho_j} \hat{h}_{ij}(k), \quad (48)$$

where ρ_i is the number density of species i .

The multicomponent OZ equations couple all correlation functions through

$$h_{ij}(r) = c_{ij}^{(2)}(r) + \sum_k \rho_k \int c_{ik}^{(2)}(|\mathbf{r} - \mathbf{r}'|) h_{kj}(r') d\mathbf{r}', \quad (49)$$

which can be expressed compactly in Fourier space in matrix form as

$$\hat{\mathbf{H}}(k) = [\mathbf{I} - \hat{\mathbf{C}}(k) \boldsymbol{\rho}]^{-1} \hat{\mathbf{C}}(k). \quad (50)$$

To close the OZ hierarchy, an approximate closure relation is required. Within the hypernetted-chain (HNC) approximation [13], the closure generalizes to

$$c_{ij}^{(2)}(r) = \exp[-\beta \phi_{ij}(r) + \gamma_{ij}(r)] - \gamma_{ij}(r) - 1, \quad (51)$$

where $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ denotes the indirect correlation function.

For systems with strong excluded-volume interactions, such as colloidal suspensions, the PY closure [13] is often more appropriate and can be written as

$$c_{ij}(r) = [1 + \gamma_{ij}(r)] [e^{-\beta \phi_{ij}(r)} - 1]. \quad (52)$$

The numerical strategy for estimating effective particle sizes and reference structures follows the same procedure as in the one-component case.

4. density dependence of the integrated strength

To assess the behavior of the system near the critical point, we fit the integrated strength A as a function of density using the two-parameter form

$$A(\rho) = \Delta B_2 (1 + \kappa \rho^\delta). \quad (53)$$

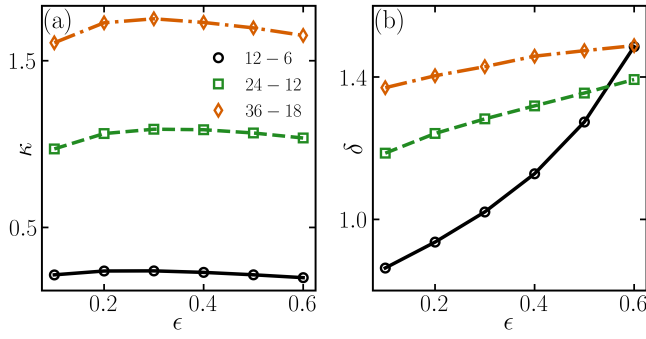


FIG. 11. Fitting parameters κ (a) and δ (b) are shown as a function of strength of interaction ϵ , for the different Mie exponent pairs as indicated in the legend, where in all the cases, we set $\sigma = 1$.

Here, the factor ΔB_2 accounts for the dilute regime. The fitting parameters κ and δ depends on the steepness of the potential (exponents of the generalized Mie potential) as shown in Fig. 11(a) and (b), respectively, for states above the critical point. Both parameters exhibit almost a monotonic variation with increasing interaction strength. This parametrization enables extrapolation of the system's behavior into the spinodal region, which is generally inaccessible through integral equation approaches or direct experimental measurements. However, this also represents a limitation of the present formalism, as the predicted behavior in this region cannot be independently verified [18].

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- [1] P. G. de Gennes, Rev. Mod. Phys. **64**, 645 (1992).
 - [2] I. W. Hamley, *Introduction to Soft Matter: Synthetic and Biological Self-Assembling Materials* (John Wiley & Sons, Chichester, UK, 2007).
 - [3] W. Poon, S. Ilett, and P. Pusey, Il Nuovo Cimento D **16**, 1127 (1994).
 - [4] W. C. K. Poon, J. Phys.: Condens. Matter **14**, R859 (2002).
 - [5] C. N. Likos, Physics Reports **348**, 267 (2001).
 - [6] C. Fick, Z. Khan, and S. Srivastava, Mater. Adv. **4**, 4665 (2023).
 - [7] Y. Zhang, Y. Luo, J. Zhao, W. Zheng, J. Zhan, H. Zheng, and F. Luo, Acta Pharm. Sin. B. **14**, 110 (2024).
 - [8] H. Wennerström and D. F. Evans, *The colloidal domain: where physics, chemistry, biology, and technology meet* (John Wiley & Sons, 2026).
 - [9] S. F. Banani, H. O. Lee, A. A. Hyman, and M. K. Rosen, Nature Reviews Molecular Cell Biology **18**, 285 (2017).
 - [10] J. M. Brader, R. Evans, and M. Schmidt, Mol. Phys. **101**, 3349 (2003).
 - [11] R. H. Nilson and S. K. Griffiths, J. Chem. Phys. **125** (2006).
 - [12] F. Llovel, A. Galindo, F. J. Blas, and G. Jackson, J. Chem. Phys. **133**, 024704 (2010).
 - [13] J.-P. Hansen and I. R. McDonald, *Theory of simple liquids: with applications to soft matter* (Academic press, 2013).
 - [14] R. Evans, Adv. Phys. **28**, 143 (1979).
 - [15] R. Evans, in *Fundamentals of Inhomogeneous Fluids*, edited by D. Henderson (Marcel Dekker, New York, 1992) Chap. 3, pp. 85–176.
 - [16] R. Evans and W. Schirmacher, J. Phys. C: Solid State Phys. **11**, 2437 (1978).
 - [17] T. Ramakrishnan and M. Yussouff, Phys. Rev. B **19**, 2775 (1979).
 - [18] J. F. Lutsko, J. Chem. Phys. **127**, 054701 (2007).
 - [19] T. C. Moore, C. R. Iacovella, and C. McCabe, J. Chem. Phys. **140** (2014).
 - [20] J. K. Percus and G. J. Yevick, Phys. Rev. **110**, 1 (1958).
 - [21] M. Wertheim, J. Math. Phys. **5**, 643 (1964).
 - [22] G. A. Mansoori, N. F. Carnahan, K. E. Starling, and T. W. Leland, J. Chem. Phys. **54**, 1523 (1971).
 - [23] J. A. Barker and D. Henderson, J. Chem. Phys. **47**, 4714 (1967).
 - [24] J. D. Weeks, D. Chandler, and H. C. Andersen, J. Chem. Phys. **54**, 5237 (1971).
 - [25] J. K. Percus and G. J. Yevick, Phys. Rev. **110**, 1 (1958).
 - [26] I. Pihlajamaa and L. M. Janssen, Phys. Rev. E **110**, 044608 (2024).
 - [27] M. Hanke, J. Stat. Phys. **170**, 536 (2018).
 - [28] H. Hansen-Guus and R. Roth, J. Chem. Phys. **124** (2006).
 - [29] Y. Rosenfeld, Phys. Rev. Lett. **63**, 980 (1989).
 - [30] R. Roth, J. Phys.: Condens. Matter **22**, 063102 (2010).
 - [31] A. J. Archer, B. Chacko, and R. Evans, J. Chem. Phys. **147**, 034501 (2017).
 - [32] A. Louis, P. Bolhuis, J. Hansen, and E. Meijer, Phys. Rev. Lett. **85**, 2522 – 2525 (2000).
 - [33] P. Bolhuis, A. Louis, J. Hansen, and E. Meijer, J. Chem. Phys. **114**, 4296 (2001).
 - [34] I. Götze, H. Harreis, and C. Likos, Journal of Chemical Physics **120**, 7761 – 7771 (2004).
 - [35] I. O. Götze, A. J. Archer, and C. N. Likos, J. Chem. Phys. **124**, 084901 (2006).
 - [36] A. J. Archer and R. Evans, Phys. Rev. E **64**, 041501 (2001).
 - [37] A. Scacchi, S. J. Nikkhah, M. Sammalkorpi, and T. Ala-Nissila, Phys. Rev. Res. **3**, L022008 (2021).
 - [38] A. Scacchi, M. Sammalkorpi, and T. Ala-Nissila, J. Chem. Phys. **155**, 014904 (2021).
 - [39] A. Scacchi, C. Rigoni, M. Haataja, J. V. Timonen, and M. Sammalkorpi, J. Colloid Interface Sci. **686**, 1135 (2025).
 - [40] V. A. Varma and A. Scacchi, Phys. Rev. E **113**, 065414 (2026).
 - [41] M. Camargo and C. N. Likos, Mol. Phys. **109**, 1121 (2011).
 - [42] A. A. Louis, P. G. Bolhuis, and J. P. Hansen, Phys. Rev. E **62**, 7961 (2000).
 - [43] F. H. Stillinger, J. Chem. Phys. **65**, 3968 (1976).
 - [44] A. Archer, C. Likos, and R. Evans, J. Phys. Condens. Matter **14**, 12031 (2002).
 - [45] A. Archer, A. M. Rucklidge, and E. Knobloch, Phys. Rev. E **92**, 012324 (2015).

- [46] B. D’Aguanno and R. Klein, J. Chem. Soc. Faraday Trans. **87**, 379 (1991).
- [47] M. Ludwig, P. Ritzert, R. Geisler, S. Prévost, and R. von Klitzing, Frontiers in Soft Matter **2**, 890415 (2022).
- [48] Y. Wang, W. Wang, K. Hong, C. Do, and W.-R. Chen, Polymer **204**, 122698 (2020).
- [49] E. Janai, A. P. Cohen, A. V. Butenko, A. B. Schofield, M. Schultz, and E. Sloutskin, Scientific Reports **6**, 28578 (2016).
- [50] J. Hirschfelder, F. McClure, and I. Weeks, J. Chem. Phys. **10**, 201 (1942).
- [51] H. N. W. Lekkerkerker, W. C.-K. Poon, P. N. Pusey, A. Stroobants, and P. B. Warren, Europhysics Letters **20**, 559 (1992).
- [52] R. Roth, R. Evans, A. Lang, and G. Kahl, J. Phys.: Condens. Matter **14**, 12063 (2002).
- [53] G. Mie, Annalen der Physik **316**, 657 (1903).
- [54] C. Avendano, T. Lafitte, A. Galindo, C. S. Adjiman, G. Jackson, and E. A. Muller, The Journal of Physical Chemistry B **115**, 11154 (2011).
- [55] A. Brünger, C. L. Brooks III, and M. Karplus, Chemical physics letters **105**, 495 (1984).
- [56] B. Dünweg and W. Paul, International Journal of Modern Physics C: Computational Physics & Physical Computation **2**, 817 (1991).
- [57] A. Z. Panagiotopoulos, Mol. Phys. **61**, 813 (1987).
- [58] R. Baxter, J. Chem. Phys. **41**, 553 (1964).